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Effect of Humidity on Photovoltaic Performance Based on Experimental Study

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Abstract
This paper presents the impact of relative humidity on the output of a solar Photovoltaic (PV). The relative humidity has influences on the other climate parameter as well as it is affected by them. So, air temperature, solar intensity and wind velocity data was collected in Sohar city-Oman and compared with relative humidity for the period from July to September 2015. The results showed that relative humidity for the tested period is highly affecting the PV performance. The PV's current, voltage and power output are decreased when the relative humidity increased. The correlation factor$R$ found to be negative between humidity and wind speed, air temperature, solar radiation these are-0.6189,-0.9764 and-0.9681, respectively. In addition, humidity coefficient of determinationR^2 is found to be strong with respect to current, voltage and power. However, the effect on efficiency is in the middle of strong and weak correlation.The results imply that the PV panel efficiency is low during high relative humidity period.

Keywords: solar energy; humidity; photovoltaic performance; efficiency.

Introduction

An increasing interest among developed and developing countries in photovoltaic (PV) has risen due to its advantages of converting solar energy into electricity on environment and economics. The climbing energy costs and growing concern about climate change generate a rapid growth in the PV industries in recent years. Although the PV efficiency is important, however, the reliability of the cell is also crucial. The PV cells subject to extreme climate conditions, as well as, their lifetime have an impact on the price per kWh determination depending on its low maintenance and operating cost [1, 2]. There are several environmental factors such as dust, humidity, temperature and air velocity influence the PV performance; in addition to, the material and design parameters. These parameters can affect the PV cell's performance. There are many studies conducted on the impact of various environmental parameters on the PV cells performance and efficiency [3, 4].

Out of 100% energy coming from the sun approximately 30% of the energy is either reflected back or absorbed by clouds, oceans, and land masses. In cities where the humidity is high, like Karachi Mumbai, Malaga, Hamburg, Muscat and Los Angles, its average ranges between (40-78 %). The Solar energy that strikes the solar cell subjected to a loss in the energy absorption/reflection. These losses are approximately of about 15-30% of the energy in addition to the mentioned30% [5].Bhattacharya [6] proposed an equation to measure the efficiency of the PV module instantly if some of the climatic parameters of specific areas are known. The measured and calculated values of efficiency are verified to check the percentage error between them. Many articles observed that developed equation has a correlation with the measured values. Temperature, humidity, wind speed and radiation define the current efficiency of a PV panel at any instant. The study conducted a statistical analysis of the independent and dependent variables for a particular location. The investigation carried out by monitoring the variation of the power output of the system with ambient temperature.

Mekhilef [7] measured the impact of humidity on the degradation in solar cell efficiency. The study observed that increasing wind velocity caused a reduction in the atmospheric air relative humidity that resulted in a better efficiency. However, high wind speed may cause higher dust lifting and scattering it in the surrounding of the PV modules resulting in lower cells performance. The study deduced that humidity, dust, and air velocity together affect the PV cells performance. The influences of these factors are reciprocal and must study it together to evaluate the efficiency of the PV cells. Gwandu [8] indicated that the humidity influences the sunlight irradiance level resulted in reducing the radiations subjected to the PV panel due to refraction, reflection or diffraction caused by water vapor particles. The reduction in the received solar radiation level causes non-linearly irradiance alterations. A little non-linear variation in open circuit voltage accompanied with significant linear variations in short circuit current.

Kazem [9] investigated the impact of relative humidity on the performance of the three types PV at Oman's climate. The obtained results illustrated that at low relative humidity conditions the output current, voltage, and power increase. Low relative humidity improves the solar cells performance. One of the study interesting findings is that Monocrystalline panel has the highest efficiency when relative humidity is decreased compared to Polycrystalline and Amorphous

Silicon technologies. Ndiaye [10] reviewed the PV panel performance degradation from differently presented modes. The research focused on corrosion, discoloration, delamination and breakage as the primary causes of the PV performance degradation. The study manifested that corrosion and discoloration are the predominant factors of PV cells degradation, although the modeling of different degradation types is still poorly examined in the literature.

Panjwani [11] studied the effect of humidity ranges between (40 to 78%). The study results indicated that there is an estimated loss of about 15-30% of the PV power. Humidity brought down the utilized solar energy about 55-60% from just 70% of utilized energy. The reason for this reduction resulted from the basal layer of water vapor lied at the front of the solar cell directly facing the sun. Where the solar irradiance strikes the solar cell face suffers from a loss in absorption/reflection energy occurs. Omubo [12] investigated the relative humidity effects on the efficient electricity conversion from solar energy. The results clarified a direct proportionality between solar flux, output current and efficiency of the solar cell. The solar flux affected by relative humidity and both have a negative impact on the PV cell output voltage. The maximum achieved PV power was at an operating temperature of 43ºC and relative humidity of about 77%.

In the recent paper, the study focuses on humidity impact on PV cells and its relation to other air parameters like temperature, wind velocity, and dust. The study aims to show how the air humidity can affect the PV performers in Oman and nearby countries.

The affected region

Sohar is one of the North Al-Batinah provinces in the Al-Batinah region in the north of Sultanate of Oman. It located 234 km north the capital Muscat. It was the ancient capital of Oman. Sohar city depends on fishing, trading, and agriculture. Nowadays, it represents the Omani industry axis due to the fast development in Sohar industrial area and port. Sohar considered as the executive center for North Al-Batenah region. It locates on the low sandy coast of Oman Sea [13].

Sohar province geographical location as a part of Oman has warm winter and hot summer seasons climate, with low rain rate. Some places in the province especially those located in the Al-Hajar Al-Garbi chain have a more moderate climate. The relative humidity is high on the coastline in the summer season especially from May to September. Sohar's relative humidity becomes low in the other months of the year. The period from October to March is suitable for truism and enjoying Sohar wonderful landmarks and views. Fig. 1 represents the geographical location of Sohar City. Table 1 indicates the climatic data for Sohar for the period from 1980 to 1990 [14].

TABLE.1.Climate data for Sohar city [15& 16]

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Record high ºC	32.6	32.1	37.4	44.5	46.9	48.5	50	45	43.2	42.4	37.7	33.9
Average high ºC	24.2	25.3	27.5	31.9	36.3	36.9	36.2	34.7	34.1	33	29.5	26.1
Daily mean ºC	18.8	19.5	22.4	26.8	31	32.7	33	31.6	30.3	27.4	23.7	20.4
Average low ºC	12.4	13.3	16.1	19.6	23.7	26.2	28.2	27.1	24.7	20.4	16.8	14.4
Record low ºC	5.7	5.8	6.8	11.2	16	19.7	22.4	21.4	17.4	12	8	7.4
Average Precipitation (mm)	4.7	56.2	17	7.8	2.5	0	0.1	0	0.5	0	3.8	15.6
Average RH (%)	72	74	72	65	63	70	77	80	79	73	72	74
Mean monthly sunshine hours	269.4	228.6	230.8	276.0	322.4	310.9	281.5	275.6	276.3	284.6	257.5	259.8

Fig. 1, Sohar city geographical location

Experimental setup

Since the 2012's the Solar Cell and Photovoltaic Lab, Engineering Faculty of Sohar University has been collecting environmental data for weather parameters such as air temperature, wind velocity and speed, solar irradiance, and relative humidity. The study has to relate the other climate conditions such as temperature, wind velocity and irradiance with humidity. So, for this purpose, a weather station type Delta-T mounted on the roof of the lab was used. The used automatic weather station is simple and suitable to supply all the application requires. The station records air temperature, relative humidity, wind speed and direction, and solar irradiance 24 hours a day. The system is dependable, rugged and suitable for research applications in demanding environments. The system includes the following: DL2e Logger (64k readings in memory), M2 Mast, AN4 Standard Anemometer, WD4 standard Wind vane, RG2 Compact Rain gauge, RG-BP Base plate for the RG2, ES2 SolarEnergy Sensor, ST1 Soil Temperature Probe and the RHT2nl combined RH and Air Temperature Sensor. Detailed specifications are available on the company website.

The tests conducted in three months (July-Augusts and September 2015). The relative humidity considered as a dependent variable, while air temperature, wind speed, and solar radiation regarded as independent variables. The average recorded data from the weather station considered as a representative of the relationships between the weather parameters.

A Monocrystalline PV panel used to study the impact of the environmental conditions especially humidity on it. The PV panel installed 27º from the horizon facing south direction, which is the optimum tilt angle as claimed by [17]. The PV output terminals, battery, and the load connected to the input terminals of the charger controller. The current and voltage readings took directly from the screen display.

TABLE. 2. The specification of PV panel

Maximum Power (P_{max})	125W
Open circuit voltage (V_{oc})	21.8V
Short circuit current (I_{sc})	7.45V
Max. power voltage(V_{mp})	17.2V
Max. power current (I_{mp})	7.27A
Cell Temperature (T_c)	25°C
Irradiation(G_a) PV L	1000W/m² 1.43 m
PV W	0.63 m
PV Area	0.9 m^2

Results and Discussions

Sohar city features by its location on the Arab Sea that linked with the Indian Ocean. This place makes it characterizes with high humidity climate. Fig. 2 represents the average relative humidity for the studied months distributed on the day hours starting at 7 AM. Relative humidity controlled by many factors one of them is the daytime. Relative humidity decreases at morning while it increases at night. However, RH in all cases does not reduce from 74%. July recorded the lowest and September recorded the highest percentage humidity.

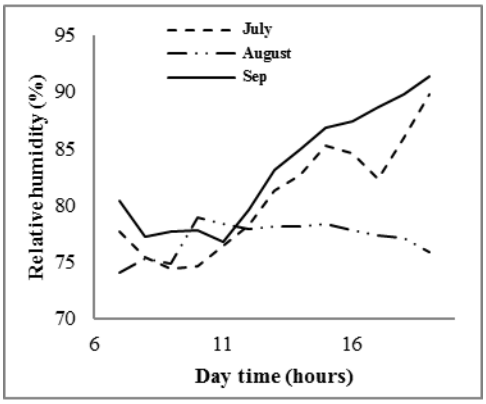

Fig. 2, average relative humidity with day time

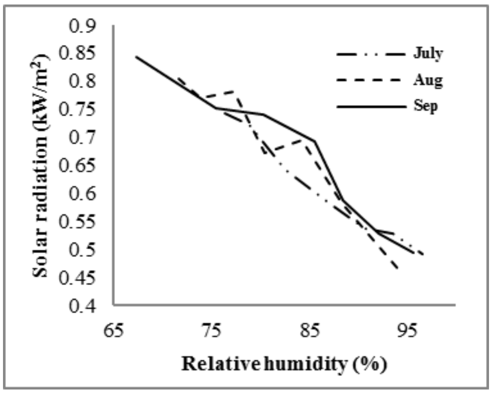

Fig. 3, the relationship between relative humidity with and solar radiation

Solar radiation reduces with increasing RH, as Fig. 3 shows. Increasing air water content increases solar radiation scattering and reduces the received solar intensity. A water vapor layer formed on the face of the solar cell in front of the sun radiation. As a result, the solar irradiance falling on the solar cell face suffers from the dispersion, and high loss in the incoming energy occurs.

Sohar characterized by its medium temperatures that it ranges from 29 to 34°C. Fig. 4 shows the relation between RH and air temperature for the studied period. RH increase causes a relative reduction in air temperature. The recorded air temperatures in Sohar summertime were suitable for the PV arrays operation. However, the RH seems as the dominated affecting factor in this city that reduces the PV performance.

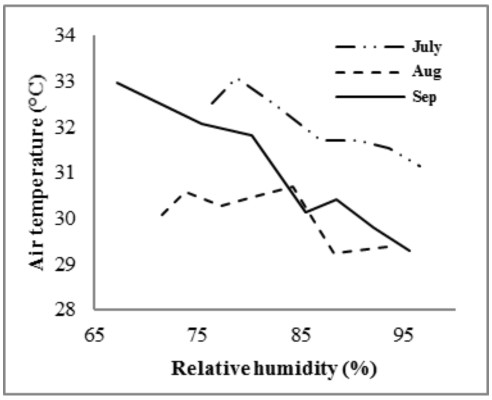

Fig. 4, the relationship between relative humidity and air temperature

The type of correlation between humidity and other parameters (air temperature and solar radiation) influences confirmed by evaluating a correlation between the measured values of these parameters. The correlation between (humidity and air temperature) and (humidity and wind speed) calculated employing MATLAB by applying the Pearson's correlation operator $\rho_{X,Y}$ as follows:

$$\rho_{X,Y} = \frac{E[(X-\mu_X)(Y-\mu_Y)]}{\sigma_X \sigma_Y} \qquad (1)$$

Where X and Y are the two variables, μ_X is the mean of X, E is the expectation, σ_X is the standard deviation of X. MATLAB used to calculate the correlation coefficients.

The correlation coefficient is ranging between-1.0 and +1.0. The positive coefficient means a positive relationship between the two variables and verse versa for the negative coefficient. The correlation coefficients between humidity and wind speed, air temperature, solar radiation are-0.6189,-0.9764 and-0.9681 respectively. This mean that the correlation is negative and the correlation with wind speed is weaker (rigorously considered as weak correlated) than that with air temperature and solar radiation.

The PV panels always expose to weather conditions, so they are subjected to wind all the time. The wind has a positive influence on the PV arrays that is cooling it by increasing the natural and forced convection. However, it has a negative impact also; it spreads the sand and dust in the air that accumulates on the surface of the PV panel after wind's cessation. Air movement reduces the RH for a certain limit, as Fig. 5 declares. The wind velocity for the studied period limited between 0 to 4 m/sec, which is a smooth, gentle air movement. The wind velocity effect in Sohar City case was limited due to the high RH rates.

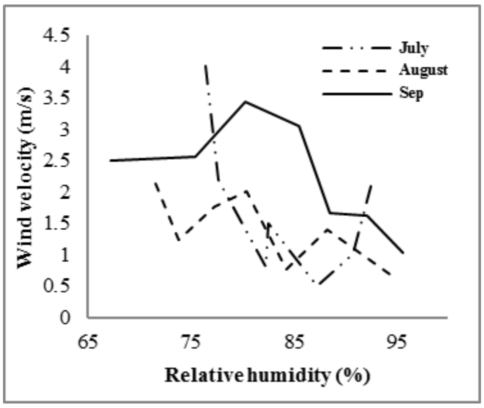

Fig. 5, the relationship between relative humidity and wind velocity

The previous figures connected the RH with other environmental parameters to clarify its effect on PV panel performance. Fig. 6 illustrates the effect of RH on the PV current produced. Increasing RH reduced current highly. Increasing RH from 67 to 95% reduced the current by 44.44%. In spite of high RH of Sohar city the PV panel produced 62% of the maximum current in the worst condition.

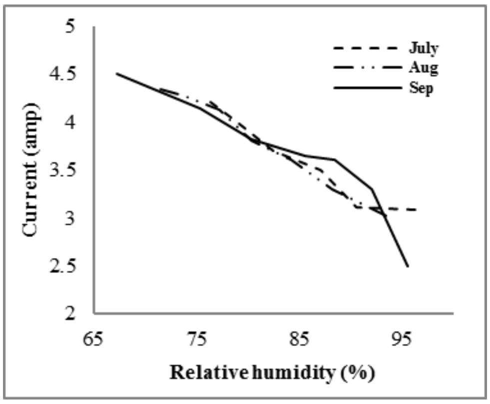

Fig. 6, the relationship between relative humidity and PV current

The voltage of PV reduced with increasing RH, as Fig. 7 represents. The RH has an adverse impact on solar radiation so that the resultant negative influence reflects on the PV cell output voltage. The PV panel voltage was reduced by 23.18, 24.88 and 25% for July, August, and September respectively.

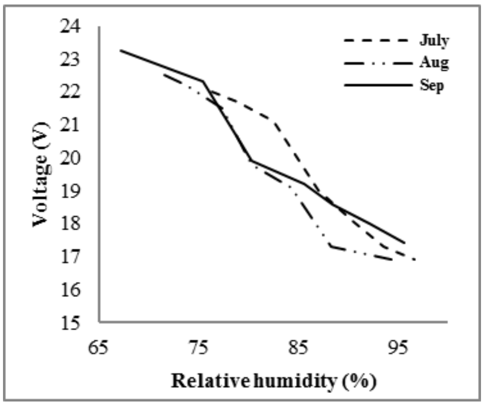

Fig. 7, the relationship between relative humidity and PV voltage

The PV array power is the result of multiplying the current by the voltage. Also, the efficiency equal to the power of PV panel / (PV area×1000W.m^{-2}).Fig. 8 shows the deterioration in the PV power with the RH increment. The resulted power reduced by 43.48, 48.42 and 58.38% for July, August, and September respectively, so that September is the worst.

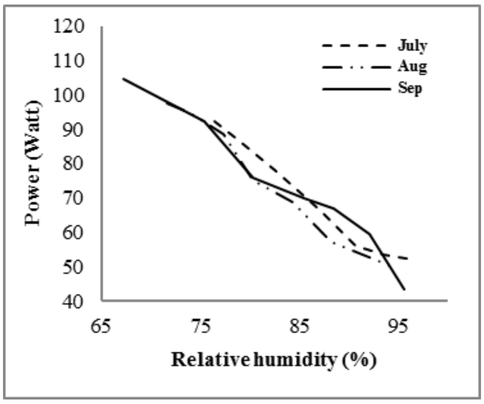

Fig. 8, the relationship between relative humidity and PV power

RH causes a reduction in the solar intensity that reduced the resulted efficiency of a PV panel, as Fig. 9 reveals. The reductions in the tested PV panel efficiency were 18.8, 19.94 and 28.77% for July, August, and September respectively. The high-efficiency reduction in September compared to the other months was due to the RH reduction lower than 70% for this month.

In order to evaluate the correlation of the humidity with current, voltage, power and efficiency, the coefficient of determination R^2 used. The coefficient of determination R^2 defined in equation 2:

$$R^2 = 1 - \frac{\sum_i (y_i - f_i)^2}{\sum_i (y_i - f\bar{y}_i)^2} \tag{2}$$

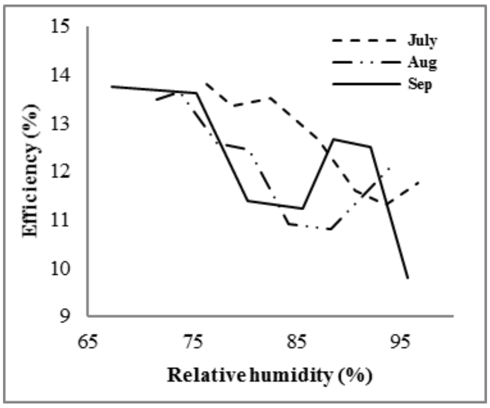

Fig. 9, the relationship between relative humidity and PV efficiency

Where $y_i$the observed value of the actual output, f_i is the predicted value and $\bar{y}_i$ is the arithmetic mean value of the observed targets. The firm correlation between variables, which achieved by the value of R^2 is closer to 1. Table 3 shows the correlation of humidity and electrical variables with R&R^2.

TABLE. 3. Humidity and electrical variables with R&R^2

Humidity %	Current (A)	Voltage (V)	Power (W)	Efficiency %
67.28	4.50	23.23	104.54	13.76
75.42	4.14	22.30	92.322	13.62
80.33	3.82	19.90	76.02	11.40
85.56	3.65	19.20	70.08	11.24
88.47	3.60	18.60	66.96	12.65
92.11	3.30	18.00	59.40	12.49
95.59	2.50	17.40	43.50	9.80
Correlation Factor R	-0.93639	-0.98245	-0.98395	-0.71935
Determination coefficient R^2	0.876828	0.965215	0.96816	0.517464

The determination coefficient achieved a value of 0.876828, 0.965215, 0.96816, and 0.517464 for the R^2 (for current, voltage, power and efficiency, respectively) and the prediction trend-line model calculated as in equations 3-6:

(x: Humidity and y: current)

$$y = -0.00022261x^3 + 0.052677x^2 - 4.174x + 114.76 \tag{3}$$

(x: Humidity and y: voltage)

$$y = 0.00032301x^3 - 0.077635x^2 + 5.9458x - 123.57 \tag{4}$$

(x: Humidity and y: power)

$$y = 4.315*10^{-5}x^5 - 0.018632x^4 + 3.1959x^3 - 272.27x^2 + 11519x - 1.9349*10^5 \tag{5}$$

(x: Humidity and y: efficiency)

$$y = -0.000226595x^4 + 0.086579x^3 - 10.513x^2 + 564.16x - 11272 \tag{6}$$

From the values of R^2 in Table 3, it is found that the humidity has a strong correlation with current, voltage and power of the PV panel. The humidity has more effect on voltage than current as the table declares. However, the effect on efficiency is in the middle of strong and weak correlation.

Conclusions

The analysis of the environmental data supplied by the weather station in Sohar University gives many significant conclusions noticed:

1- RH changes with time, and it is higher at night hours than at daytime. The main result is that the daytime expands in Sohar summer times for more than 12 hours.

2- The air water vapour contents affect the solar irradiance level and reduce the solar intensity by scattering the radiation.

3- RH in Sohar city is the dominant parameter affecting the PV performance as the air temperature in this city is moderate and does not exceed 35°C.

4- Wind speed in Sohar city is low. As a result, its influence in reducing the RH is limited.

The relative humidity causes degradation in solar cell current, voltage, power, and efficiency. The study confirms the high impact of RH in this degradation compared with the other studied parameters in Sohar city. The long period of daytime during the Sohar summertime, high solar radiation and high maximum peak hours makes it useful to use solar PV panels in spite of the high relative humidity.The correlation factor R and determination R^2coefficient of the humidity with current, voltage, power and efficiency, has been calculated. It is found that humidity in an inverse strong correlation with current, voltage and power. However, the humidity correlation and effect is less on PV efficiency.

References

[1] Koehl M, Heck M and Wiesmeier S, Modeling of conditions for accelerated life time testing of Humidity impact on PV-modules based on monitoring of climatic data, Solar Energy Materials & Solar Cells 99 (2012) 282-291

[2] Elminir H K, Benda V and Tousek J, Effects of solar irradiation conditions and other factors on the outdoor performance of photovoltaic modules, Journal of Electrical Engineering, vol. 52, No. 5-6, pp: 125-133, 2001.

[3] Kempe M D, Modeling of rates of moisture ingress into photovoltaic modules, Solar Energy Materials and Solar Cells, vol. 90, No. 16, pp: 2720-2738, 2006.

[4] Koehl M, Heck M and Wiesmeier S, Modeling of conditions for accelerated life time testing of Humidity impact on PV-modules based on monitoring of climatic data, Solar Energy Materials & Solar Cells 99 (2012) 282-291

[5] Han X, Wang Y, Zhu L, Xiang H and Zhang H, Mechanism study of the electrical performance change of silicon concentrator solar cells immersed in de-ionized water, Energy Conversion and Management, vol. 53, No. 1, pp: 1-10, 2012.

[6] T. Bhattacharya,A. K. Chakraborty, and K. Pal. " Statistical Analysis of the Performance of Solar Photovoltaic Module with the Influence of Different Meteorological Parameters in Tripura, India".International Journal of Engineering Research. ISSN: 2319-6890)(online),2347-5013(print). Volume No.4, Issue No.3, pp: 137-140 01 March. 2015.

[7] Mekhilef S, Saidur R, Kamalisarvestani M, Effect of dust, humidity and air velocity on efficiency of photovoltaic cells, Renewable and Sustainable Energy Reviews, vol. 16, pp: 2920-2925, 2012.

[8] Gwandu B A L, Creasey D J, Humidity: a factor in the appropriate positioning of a photovoltaic power station, Renewable Energy, vol. 6, No. 3, pp: 313-316, 1995.

[9] Kazem H A, Chaichan M T, Al-Shezawi I M, Al-Saidi H S, Al-Rubkhi H S, Al-Sinani J K and Al-Waeli A H A, Effect of Humidity on the PV Performance in Oman, Asian Transactions on Engineering, vol. 2, No. 4, pp: 29-32, 2012.

[10] Ndiaye A, Charki A, Kobi A, Ke´be´ C M F, Ndiaye P A, Sambou V, Degradations of silicon photovoltaic modules: A literature review, Solar Energy, vol. 96, pp: 140-151, 2013.

[11] Panjwani M K, NarejoG B, Effect of humidity on the efficiency of solar cell (photovoltaic), International Journal of Engineering Research and General Science, vol. 2, No. 4, 2014.

[12] Omubo-Pepple V B, Israel-Cookey C and Alamunokuma G I, Effects of temperature, solar flux and relative humidity on the efficient conversion of solar energy to electricity, European Journal of Scientific Research, vol. 35, pp: 173-180, 2009.

[13] Agius D A, Classic ships of Islam, from Mesopotamia to the Indian Ocean, IDC publisher, Najhoff publishers and VSP, 2008.

[14] Sohar Port and Freezone-Overview [http://www.soharportandfreezone.com/en/about/overview]

[15] National Oceanic and Atmospheric Administration (NOAA), Majis Climate Normals 1980-1990. (all but average maximum, 1980-1990), Retrieved January 15, 2013.

[16] Sohar climate, www.world-climates.com (average maximum), Retrieved January 15, 2013.

[17] Kazem H A, Khatib T, and Alwaeli A A, “Optimization of Photovoltaic Modules Title Angle for Oman”, 7th IEEE International Power Engineering and Optimization ConferencePEOCO2013, Malaysia, 3-4 June 2013, pp. 700-704.